

Connect to Databases

SWEN1 | BIF3

Table of Contents

- Connection Basics
- PostgreSQL Connection Parameters
- JDBC in Java (IntelliJ)
- ADO.NET in C# (Visual Studio)
- Troubleshooting



Connection Basics

PostgreSQL Connection Parameters

- Applications need to read and write data persistently
- PostgreSQL runs in a Docker container
- We connect our Java or C# app to this container
- We use:
  - **JDBC** (Java)
  - **ADO.NET** (C#)
- The database server listens on `localhost:5432`



PostgreSQL Connection Parameters

Parameter	Value
Host	`localhost`
Port	`5432`
User	`postgres`
Password	`mysecretpassword`
Database	`postgres` (default)

📌 These values must match your Docker container setup

📌 PostgreSQL must be running

📌 Docker container `swen-db` must expose port `5432`



JDBC in Java (IntelliJ)

What is JDBC?

- JDBC = Java Database Connectivity
- Standard API for connecting to SQL databases
- We use the PostgreSQL driver (`org.postgresql.Driver`)

Key classes:

- `Connection`
- `Statement` / `PreparedStatement`
- `ResultSet`

👉 JDBC sends raw SQL over TCP and handles result sets

Required Setup (Java)

✅ **JDK 21+**
✅ **IntelliJ** with Maven support
✅ PostgreSQL running via Docker

Add driver to `pom.xml`:

```xml
<dependency>
  <groupId>org.postgresql</groupId>
  <artifactId>postgresql</artifactId>
  <version>42.7.7</version>
</dependency>
```

Example: Java Connection

```
String url = "jdbc:postgresql://localhost:5432/postgres";
String user = "postgres";
String password = "mysecretpassword";

try (Connection conn = DriverManager.getConnection(url, user, password)) {
    System.out.println("Connected!");
} catch (SQLException e) {
    e.printStackTrace();
}
```

📌 Use import java.sql.*;



ADO.NET in C# (Visual Studio)

What is ADO.NET?

- ADO.NET = Low-level database API for .NET
- Part of .NET Base Class Library
- We use the Npgsql provider for PostgreSQL

Begriff Bedeutung

- ADO.NET The general .NET framework for database access
- Npgsql A specific ADO.NET driver for PostgreSQL

Key classes:

- NpgsqlConnection
- NpgsqlCommand
- NpgsqlDataReader

👉 ADO.NET executes SQL and reads results directly

Required Setup (C#)

✅ .NET 8+
✅ Visual Studio 2022
✅ PostgreSQL running via Docker

Install Npgsql via NuGet:

- Open NuGet Manager
- Search for Npgsql
- Install it

Or use CLI:

```
dotnet add package Npgsql
```

Example: C# Connection

```
var connString = "Host=localhost;Port=5432;Username=postgres;Password=mysecretpassword;Database=postgres";

using var conn = new NpgsqlConnection(connString);
conn.Open();
Console.WriteLine("Connected to PostgreSQL!");
```

📌 Use using Npgsql;



Troubleshooting

Common Errors & Fixes

Error	Cause
`Connection refused`	Container not running
`Timeout expired`	Port/firewall problem
`Authentication failed`	Wrong password or user
`No suitable driver` (Java)	Missing Maven dependency
`ConnectionString error` (C#)	Missing keyword/value

Summary

✅ You can now connect to PostgreSQL
✅ You use JDBC (Java) and ADO.NET (C#)
✅ You verified Docker and connection status

Further Reading

- (JAVA) JDBC Documentation (PostgreSQL)
- (C#) Npgsql (ADO.NET for PostgreSQL) Documentation
- PostgreSQL psql Command-Line Tool